

Java ArrayList – 10 Coding Questions

Practice / Assignment

1. Remove Duplicates

Write a program to remove duplicate elements from an ArrayList while maintaining the original order.

Test Case:

Input: [10, 20, 10, 30, 20, 40]

Output: [10, 20, 30, 40]

2. Second Largest

Find the second-largest distinct element in an ArrayList.

Test Case:

Input: [10, 50, 20, 50, 30, 40]

Output: 40

3. Frequency of Elements

Find the frequency of each element in an ArrayList.

Test Case:

Input: [10, 20, 10, 30, 20, 10]

Output:

10 -> 3

20 -> 2

30 -> 1

4. Common Elements

Given two ArrayList<Integer> objects, find the elements common to both.

Test Case:

Input:

List1 = [10, 20, 30, 40, 50]

List2 = [30, 40, 60, 70]

Output: [30, 40]

5. Separate Even and Odd

Create two ArrayLists containing even and odd numbers separately.

Test Case:

Input: [11, 20, 31, 42, 53, 64]

Output:

Even: [20, 42, 64]

Odd: [11, 31, 53]

6. Reverse Without Collections.reverse()

Reverse an ArrayList without using Collections.reverse().

Test Case:

Input: [10, 20, 30, 40, 50]

Output: [50, 40, 30, 20, 10]

7. Find Missing Number

An ArrayList contains numbers from 1 to n, with one number missing. Find the missing number.

Test Case:

Input: [1, 2, 3, 5, 6, 7]

Output: 4

8. Merge and Remove Duplicates

Merge two ArrayLists and remove duplicate elements.

Test Case:

Input:

List1 = [1, 2, 3, 4]

List2 = [3, 4, 5, 6]

Output: [1, 2, 3, 4, 5, 6]

9. Rotate ArrayList

Rotate an ArrayList to the right by k positions.

Test Case:

Input: [1, 2, 3, 4, 5], k = 2

Output: [4, 5, 1, 2, 3]

10. Find Largest and Smallest

Find both the largest and smallest elements without using Collections.max() or Collections.min().

Test Case:

Input: [45, 12, 78, 23, 9, 56]

Output:

Largest = 78

Smallest = 9